

Sustainable Energy Production through Electrolyzed Water Using Solar-Powered Hydrogen Generation

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Abstract - This project explores the viability of hydrogen production via solar-powered electrolysis and its use as a clean energy carrier. It outlines the thermodynamics of water electrolysis, evaluates solar integration, and compares energy recovery through hydrogen fuel cells and combustion. The study highlights system efficiency, and practical limitations, offering insight into the potential for sustainable hydrogen-based energy systems and improvements that can be made in existing methods for better energy extraction. Solar energy production was used as the primary energy source for electrolysis, where it was found that the most efficient solar – electrical energy conversion panels were 25%. Two methods of electrolysis were studied AWE and PEM, with the inspection of a gyroid electrode, which performed with over a factor 3x in relation to regular AWE electrodes. PEM electrolysis had the highest energy efficiency (180%) through combustion energy recovery of hydrogen. The other methods of hydrogen energy recovery researched were mechanical energy through combustion which had an energy output efficiency of 55%, and hydrogen fuel cells for electrical energy generation (65%). The overall energy efficiency of green hydrogen was analyzed and the best path for hydrogen energy generation was determined to be combustion of H₂ gas produced from PEM electrolysis. The gyroid AWE also performed similarly to PEM in terms of hydrogen production rates, which is a better alternative to commercially used AWE Ni electrodes.

Keywords – PEM (Proton exchange mechanism), AWE (Alkali water electrolysis), PEMFC (Proton Exchange Membrane Fuel Cell)

energy conversion by reducing overall energy wastage in contrast to fossil fuels.

1.3 Overview of water electrolysis and solar power integration

1. Introduction

1.1 Background on global energy challenges and the role of renewables

As civilization advances so does their need for energy and while high energy production technology such as nuclear fission reactors exists, complex geo-politics due to the weaponization potential of fissionable material, the complex technical requirements for operation and the catastrophic consequences due to reactor failure prevent widespread usage of this technology (US Department of Energy, 2024). This has left a high dependency on fossil fuels for power generation as many renewable energy sources do not adequately substitute the high energy density of the chemical energy released by hydrocarbons (Ritchie et al., 2024). This makes hydrogen vital as it has a higher chemical potential energy than hydrocarbons and energy can be extracted other than through combustion due to recent advances in hydrogen fuel cells.

1.2 Importance of hydrogen as a sustainable fuel

Hydrogen has many advantages over conventional fuels mainly its high energy density lack of harmful emissions makes an ideal transition from fossil fuels (Ji, M. and Wang, J., 2021). Not only does it have over twice the energy density of other fuels but in recent years the development of hydrogen fuel cells makes its usage much more versatile as electricity can be directly produced from hydrogen gas thereby eliminating the need for multi-step

Much of the hydrogen produced globally comes from non-renewable sources (IRENA, 2022) only around 4% comes from renewable sources which makes the use of hydrogen as a green fuel/ energy source futile, as the emissions produced during the production of hydrogen negates its overall environmental benefits. While water electrolysis is one of the methods being used commercially the power for this does come from the grid which is mainly hydrocarbon power. Water electrolysis does stand out due simplicity, ease of set up and is much energy efficient than other methods (Wikipedia Contributors, 2019). Therefore, using a consistent renewable energy source for production is vital for fully sustainable hydrogen power.

Both solar and wind power are good contenders and while however wind power is more efficient at producing electricity, it's extremely location dependent, power generation is quite variable and setup is much more complex than that of solar panels (Benn, T., 2023). Therefore, solar panels are much more viable for long-term consistent commercial hydrogen generation.

1.4 Research aims and objectives

This project aims to find the most optimal ways to produce energy through hydrogen and it considers a variety of factors from efficient solar power generation, best electrolysis method, efficient energy extraction via hydrogen and environmental impact. This study mainly focuses on solar power as it's the most green and

consistent power source available but its not without its drawbacks and according to the laws of thermodynamics when energy is transferred from one form to another there is loss that is why it's important to find the most energy efficient route for solar power generation, electrolysis and hydrogen energy generation. This will give the way to create hydrogen with zero net emissions while creating a fuel source that could potentially replace fossil fuels.

2. Literature Review

2.1 Solar Energy Systems

Solar energy is most abundant form of renewable energy available all over the world, but it tends to vary and dependent on factors such as solar irradiance (Akpınar et al., 2025) shows significant improvement in power output by introducing solar tracking technology which has increased by 22.7% in electricity and 25.6% in heat.

In obtaining solar energy for water electrolysis there are 2 main systems that are being used mainly, Photoelectrochemical Cell (PEC) and Photovoltaic-Electrolyser with Voltage Converter (PE-VC) systems. According to (Qureshi, F. and Tahir, M., 2024) "The efficiency trend indicates that PV-EC systems have the highest STH efficiency (32%), preceded by PEC (19%) and PC systems and solely PE-VC systems can ensure over a decade of long-term durability for feasible solar production of hydrogen." Therefore, for this study PE-VC systems were considered. However, they did cover many improvements in PEC systems which should be theoretically more efficient as PEC directly converts solar radiation into hydrogen while PE-VC systems use a much more multistep approach of using electricity generated by solar panels passed through an electrolyser for hydrogen production. This disparity seen in PECs is due to the relatively new use of this technology and the low energy density of photovoltaic generation.

There has been quite a large scientific endeavor to negate these drawbacks in PECs as they do have promise to replace PE-VCs as mentioned by (Mohd et al., 2024) does cover significant improvements made such as the introduction of nanostructures with various metal catalysts to enhance the STH but these improvements still yet to have outperform other electrolyser methods.

In PE-VC systems the power for the electrolysis process is derived externally to the electrolyser even though there are many methods on this (Shiva et al., 2022) highlights how currently nearly 95% of the hydrogen produced commercially is from non-renewable sources their impact and most importantly it is highlighted importance of using solar and wind power for hydrogen generation for net zero emissions. However, the challenges of solar and wind are not highlighted.

2.2 Electrolysis Technologies

There are several types of electrolysis methods for water electrolysis, (Vedrtnam et al., 2025) cover in depth the various types of water electrolysis and the advances made in them for much more efficient hydrogen production. Alkaline Water Electrolysis (AWE), Polymer Electrolyte

Membrane Water Electrolysis (PEMWE), Solid Oxide Electrolysis Cells (SOEC), and Microbial Electrolysis Cells (MEC) are the main types of commercially used electrolysis discussed, although all of them show promise only AWE and PEMWE electrolyzers are compatible with solar power, as SOEC systems require high operational temperatures (500 °C -1000 °C) which is non-viable with solar integration and MEC are dependent on specific micro-organisms and both these technologies are still mainly in the R&D phase as (El-Shafie, M. 2023) also points out. Furthermore, AWE and PEMWE (PEMs) are electrolyzers that can be operated via solar power as their operating conditions are close to normal environmental conditions and various operating conditions are highlighted for optimal hydrogen production.

As well as by (Emam et al., 2023) which collaborates their research with more experimental data with respect to Alkaline concentration, electrode spacing and surface area. (Hassan et al., 2024) Illustrates the structure of AWE and PEMs however it does not cover various types of conditions that can be used only as it focuses on PEC systems. (Burton, N.A. and Grant, J.C., 2025) have pointed out the effect of magnetic fields on water electrolysis as water is a dipolar molecule it shows an overall increase but the power necessary for this process might not be compensated by its STH ratio increase.

2.3 Hydrogen Utilization

Hydrogen is used in a variety of industries from transport to power generation (Hwang et al., 2023) points out how today 65% of global emissions from these sectors and 6% of the global energy is used for hydrogen production. Furthermore, it specifies that 75% of the hydrogen produced is from methane which releases carbon gas emissions (CO and CO₂). However, this study focuses on mainly on the decarbonization transport sector with hydrogen powered vehicles which is a crucial step but the focus on green hydrogen production for this is sparsely looked at. Hydrogen has a variety of uses, even aviation (Soleymani et al., 2024), points out the potential of hydrogen to serve as a viable substitute for jet fuel in jet turbines. However, this study does not explore the possibility of the use of atmospheric water vapor electrolysis for power. (Gao et al., 2025) also looks at the combustion potential of hydrogen in piston which shows it is much more energy efficient than other fuels used as stated "The high flame propagation velocity of hydrogen combustion enables neat hydrogen-fueled operation to achieve superior indicated thermal efficiency", as these engines would run much more economically for users as the even the heat wastage is much lower than conventional engines. The major challenges that lie in conversion to hydrogen powered vehicles is its production still utilizes hydrocarbons and the storage requirements which needs to be pressurised and used special delivery methods to the engines. (Wang, B. et al., 2024) also further backs this research with more investigation into the specifics into the operational factors of a hydrogen engine like spark timing, various concentration of hydrogen to oxygen, ideal cylinder and piston shapes for a hydrogen combustion engine. Furthermore, (Wang, Y. et al., 2023) has investigated a hydrogen engine prototype with simulations

which confirms the viability of hydrogen as a fuel for piston engines and the results closely mimics the theoretical calculations and CFD simulations. Hydrogen fuel cells provide an alternative path for direct electricity generation from hydrogen with minimizing thermodynamic loss through energy conversion as (Zotov, L.G. and Bolshakov, I.M., 2024) investigates existing commercial use of hydrogen fuel cells in vehicles and notes and suggestions into improvement. However, this study mainly focuses on the transport industry not the wide application of power generation. (Yan et al., 2025) investigates the power generated by PEMFC (Proton exchange membrane fuel cell) which shows that the power generated from a stack of PEMFCs can theoretically be sufficient for household power. However, this study uses ammonia decomposition for the hydrogen feed which was an insufficient quantity during certain times of the day but it was important data for the amount of hydrogen required for power generation and water splitting could be sufficient as it produces a one to one molar ratio of hydrogen in contrast to ammonia which only produces 0.5 moles of hydrogen for every mole of ammonia. (Myatezh, S.V. and Lisitsyn, P.S., 2024) investigates the conversion of the DC to AC for commercial usage as the grid operates on AC and the power generated by fuel cells is DC. Looks into the use of efficient conversion using multi-winding power transformers instead of chains of charged capacitors for smooth power with minimal wastage and the prevention of energy surges which can prolong the longevity of fuel cells.

3. Methodology

The methodology for this project will follow a step-by-step process as illustrated in (Fig. 1). Looking at a variety of variables involved with electrolysis and improvements made for existing processes to increase the overall energy efficiency of the system. All the equations, energy and hydrogen production values have been validated based on the literature review and the key improvements made in them have been taken into consideration in this project. The two electrolysis methods analyzed are PEM and AWE electrolysis. Their operational conditions are much more energy efficient, the power used will be through solar power panels, the approach will look at traditional solar input and the increase done by solar tracking. Furthermore, a gyroid shape electrode will be analyzed as well in respect to AWE electrolysis, as a key improvement to existing electrodes.

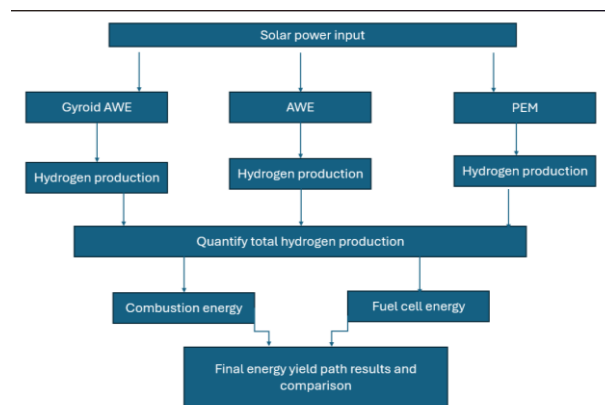


Fig. 1: Graphical methodology

As investigation suggests the best solar to electric conversion efficiency panels are ones made by a UK based company Oxford PV (McClements D., 2022) there solar cells have a conversion rate of 25% therefore these cells in conjunction with STS systems (Akpınar et al., 2025) which achieved an average power output of 1087.21 W/m² will be used in calculating the power for electrolysis.

AWE electrolyzes as demonstrated in Fig. 2 use two metal electrodes which when an electric field is passed through break the hydrogen oxygen bonds and produce hydrogen and oxygen gases. Where at the anode oxygen gas is produced while the anode releases hydrogen. Which utilizes Nickle (Ni) and Cobalt (Co) electrodes for large scale hydrogen production along with a Potassium hydroxide solution(KOH) used in 30-40% concentration for optimal hydrogen production as a catalyst for the reaction (El-Shafie, M. 2023).

The PEM electrolyzes as demonstrated in Fig. 2 uses a proton exchange membrane along with electrodes to produce hydrogen gas. These electrolyzes are standard for safe sustainable use operating at 1.80V and a current density of 1 A/cm² (Vedrtnam et al., 2025) however, the PEM electrolyzes have a few drawbacks compared to regular AWE electrolysis as the proton exchange membrane are susceptible to degradation in case of power overload or prolonged use.

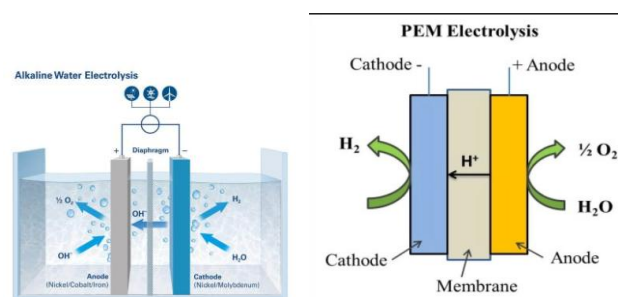


Fig. 2: AWE and PEM

The gyroid is a shape with high surface area and porosity which makes it ideal for the electrolysis process and its performance will be investigated as an alternative to

traditional flat face electrodes for AWE process. Fig. 3 has demonstrated a modelled structure of this electrode.

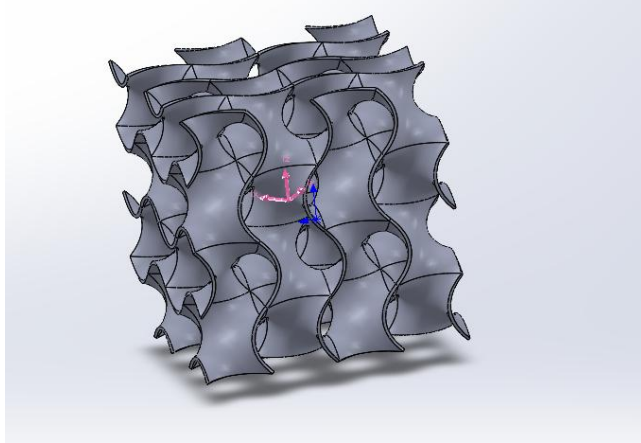


Fig. 3: Gyroid structure

This shape has much complexity and till recent advances in metal 3d printing technology it was impossible to replicate but now with advances in metal 3d printing (Gadagi, B. and Lekurwale, R., 2020) these electrodes can be used. This study uses a modelled $40 \times 40 \times 40 \text{ mm}^3$ gyroid electrode in comparison to traditional electrodes and PEM electrolyzers. Once hydrogen production is calculated from the above methods the most efficient method will be considered and for set volume/moles of hydrogen the energy produced via combustion and fuel cells will be analyzed and compared to the energy required for its production.

Hydrogen combustion technologies have been utilized in power generation and transport sectors, where it is confirmed that it has a much higher energy density of 119.7 MJ/Kg (mega joules per kilogram) compared to other fuels as shown in Table 1 (Hwang et al., 2023).

Table 1: hydrogen yield

Parameters	Hydrogen	Gasoline	Diesel	Natural Gas	Alcohol
Carbon content/%	0	84	86	75	75
Low calorific value/(MJ/kg)	119.7	44.8	42.5	45.8	45.8

Hydrogen fuel cells take in hydrogen and oxygen gas directly and use a catalytic material (typically Pt or Pd) to break the intermolecular bonds where the hydrogen is split into ionised protons (H^+) which are sent through a proton membrane and the electrons are diverted externally which creates a potential difference produces an electrical current and the subsequent protons are combined with oxygen to produce water as shown in Fig. 4. The power produced is highly dependent on the catalytic surface area requiring multiple stacks of fuel cells depending on the rate of hydrogen gas flow (Kumar, R. and Subramanian, K.A., 2022).

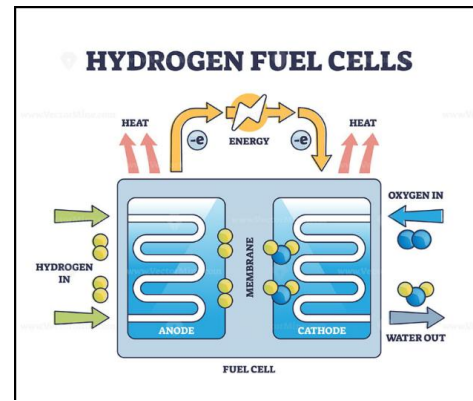
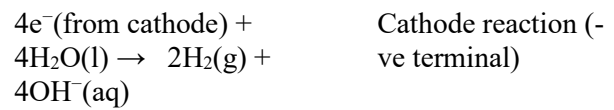
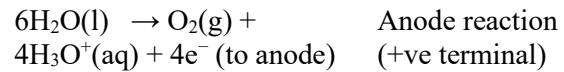


Fig. 4: Hydrogen fuel cell

3.1 Chemical equations

This section highlights the chemical process that is taken during electrolysis, both at the anode and the cathode redox reactions convert water into hydrogen and oxygen gas.

The chemical equation of electrolysis process is defined as:



Gibbs free energy is the activation energy required for a reversible thermodynamic reaction (Kamran, M., 2021). Gibbs Free Energy equation is used to determine the energy for electrolysis (ΔG , Gibbs Free Energy) for this reaction is 237.13 kJ/mol ($\Delta G = \Delta H - T\Delta S$) these values are for a reaction at room temperature "T" ($25^\circ\text{C} / 293 \text{ K}$). ΔS is the change in entropy and ΔH is the change in enthalpy for this reaction. The theoretical minimum potential difference required to electrolyze water is calculated to be 1.23V . Therefore, our solar energy system will need a minimum potential difference of 1.23V for reaction initiation. However, this is only the theoretical minimum potential difference needed to initiate the reaction to calculate the potential difference required for an optimal electrolysis an overpotential is required. Which is the voltage above the theoretical minimum to drive electrolysis and to overcome electron repulsion. This is done by the Tafel equation and the Tafel slope which gives a comprehensive understanding of the factors involved in electrolysis and their influence on the reaction.

$$(1) \quad \eta = a + b \cdot \log_{10}(j)$$

Where:

- η is the overpotential (the difference between the applied potential and the equilibrium potential, $E-E_{eq}$).
- a is related to the exchange current density (j_0), a measure of the intrinsic rate of the reaction at equilibrium. Specifically, $a=-\log(j_0)$.
- b is the Tafel slope, which provides information about the mechanism of the electrochemical reaction and its rate-determining step. It's typically expressed in mV/decade.
- j is the current density (current per unit area).

$$\eta = V_{\text{measured}} + V_{\text{thermodynamic}}$$

Eq. (1) helps in the calculation of the overpotential required for a given material for AWE process, the typical industrial process uses mainly Nickle based electrodes due to their low cost and high reaction rate. Using python, a Tafel plot for Nickle overpotential for a specific area has been simulated as seen in Fig. 5.

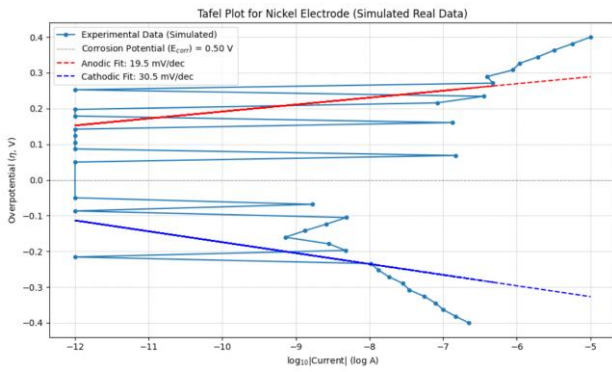


Fig. 5: Tafel plot for Nickle electrodes

Once the required Tafel plot was generated, the reaction constants (a and b) for Nickle in respect to specific current density (A/cm^2) was plotted as shown in Fig. 6.

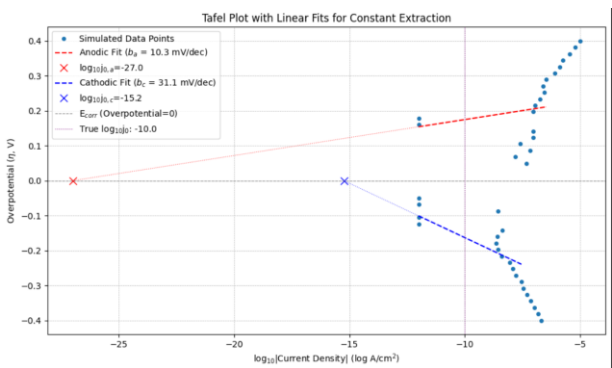


Fig. 6: Tafel constant distribution for Nickle

3.2 Energy Analysis

The power provided by solar panels have limited efficiency of conversion the best existing solar panels as discussed above has an energy conversion rate 25% with a solar energy absorption rate of 1087W. Therefore, the energy input for the electrolyzes is $1087 \times 25\% = 271.25W$. A 2V(volts) system will make use of this power

and we can calculate the current passing through the electrodes with Eq. 2.

$$(2) P = I \times V$$

Eq. (2) shows us the relationship between power, current and voltage, where the power is equal to the product of the current(I) and voltage(V). rearranging this we get, $I = P/V$ ($I = 271.25/2 = 135.875 A$). Hence the charge passed through the system can be determined in relation to eq. (3).

$$(3) Q = I \times t$$

Where Q is total charge in Coulombs (C), I is the current in Amperes (A) and t is the time in respect to seconds. Once the charge passed for a given time is calculated the number of moles of electrons passed through the electrolyzer is calculated using the Faraday constant ($F \approx 96485 C/mol$), which is the charge carried by one mole of electrons. Where F is the product of the elementary charge, C_e (charge of one electron, 1.602×10^{-19}) and Avogadro's number (which is the number of particles atom/molecules in each mole of that substance, $N_A = 6.022 \times 10^{23}$) as shown in eq. (4).

$$(4) F = N_A \times C_e$$

Using this equation for number of electrons (N_e) in respect to charge (Q) and the Faraday constant (F) as shown in eq. (5) can be extrapolated.

$$N_e = \frac{Q}{F}$$

Hence, theoretical equations under ideal conditions for the number of moles of Hydrogen(nH_2) and Oxygen(nO_2) produced was created as shown in eq. (6) and eq. (7)

$$(6) nH_2 = \frac{Q}{2F}$$

$$(7) nO_2 = \frac{Q}{4F}$$

3.3 Electrolysis calculations

For standard electrodes the shape of electrodes used are flat plate structures which we will equate to the same amount of material as the gyroid shape seen, the volume of the modelled gyroid is $4949.82mm^3$. A flat plate with 2mm thickness with the same volume will have the following dimensions as seen in table (1)

Table: 1 electrode dimensions

volume	4949.82
thickness	2
area of one side	2474.91
base	28.72229099
highet	86.16687298
surface area(mm2)	5409.376656
surface area(cm2)	54.09376656

The overall exposed surface area when submerged in the electrolyte for a flat face Ni electrode is approximately 54 cm^2 having the same volume as the modelled gyroid. Ideal operating conditions for a 2V electrode is a current density (j) of 2 A/cm^2 as highlighted by (Vedrtam et al., 2025).

For the modelled gyroid electrode the physical properties are listed in Fig.7 where the total surface area when submerged in the electrolyte is approximately 203 cm² which will have the same voltage and current density as the regular electrode.

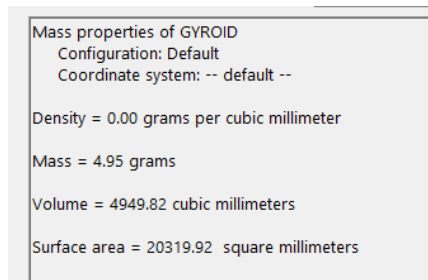


Fig.7: gyroid properties

Using these values plots for the theoretical hydrogen production rates was obtained as seen in Fig.8

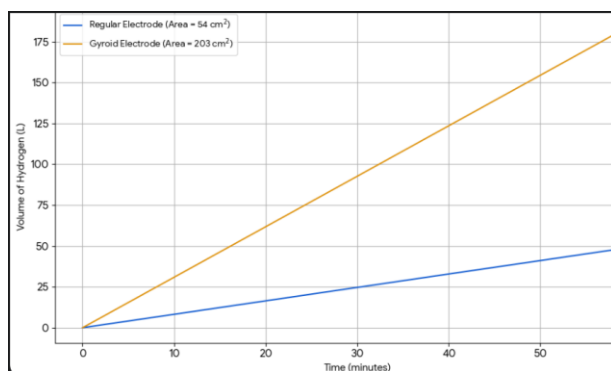


Fig. 8: gyroid and regular electrode Hydrogen production rates

This plot was for standard conditions of pressure 1 ATM (101.325 kPa) and temperature 273.15 Kelvin. The time is measured in minutes and the volume in liters. Both electrodes are immersed in a 30% KOH solution as this is the most effective as provided by Fig. 9 (b) (Emam et al., 2023). For Ni based electrodes as shown by Fig. 9 (a) the hydrogen production efficiency of nearly 80% was achieved (Vedrtnam et al., 2025).

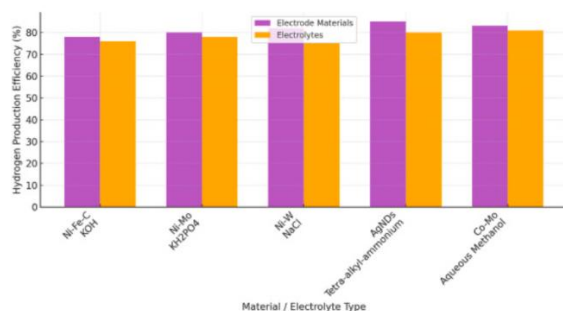


Fig. 9 (a): Electrode material/electrolyte solution efficiency (Vedrtnam et al., 2025)

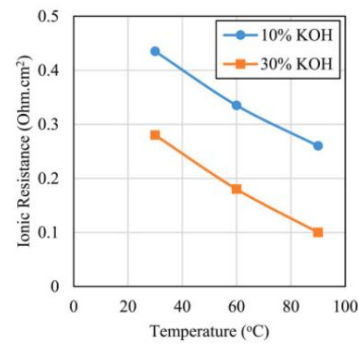


Fig. 9 (b): KOH concentration comparison (Emam et al., 2023)

When both Fig 9 (a) and Fig 9 (b) are compared a 30% KOH solution is the best fit in terms of hydrogen production and energy efficiency as the Ohmic resistance is significantly lower at room temperature, which reduces the energy loss and increases the rate of reaction due to higher availability of free ions for charge transfer between the anode and the cathode.

In PEM electrolyzers the catalytic material used is mostly Platinum (Pt) and it utilizes a membrane for hydrogen production and can only support a current density of 1 A/cm² as additional power will lead to membrane degradation. Furthermore, the electrolyser requires highly purified water for its operation as contaminants lead to membrane damage along with performance loss in hydrogen production due to larger ions present in impure water (such as Mg⁺, Ca⁺, Na⁺, Cl⁻...etc.) present in water prevents the H⁺ ion diffusion through the membrane by being embedded on the membrane wall. Therefore, the production with ultra-pure water is the most efficient method for electrolysis using PEM electrolyzers as the impurities lead to increased Ohmic resistance, thereby increasing the energy required for electrolysis as shown in Fig.10(a). (PyaePyae et al., 2024) experimental study shows the effect of introducing soft water (water with various ions) to a PEM electrolyser system which was operating with ultrapure water and after a certain time of electrolysis soft water was introduced to the system as provided in Fig.10(b).

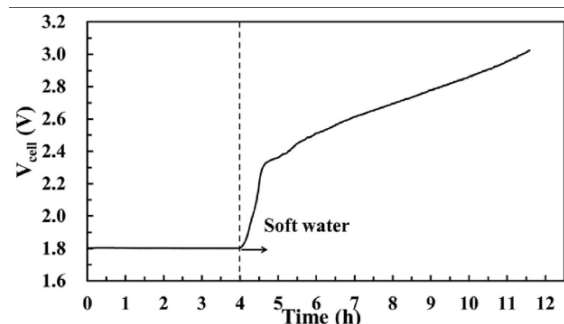


Fig. 10 (a): PEM voltage analysis with soft water introduction (PyaePyae et al., 2024).

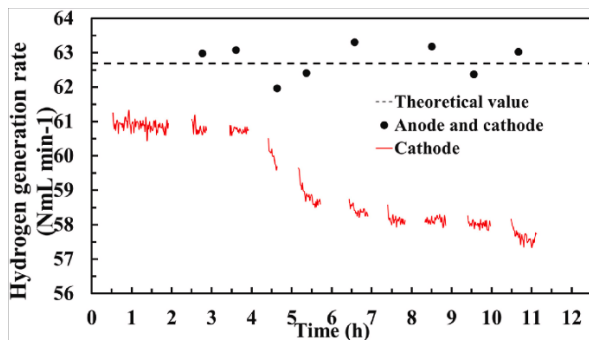


Fig. 10 (b): PEM hydrogen production rate with the introduction of soft water (PyaePyae et al., 2024).

In Fig. 10 it is observed that the PEM performance drastically changes with the introduction of impurities to the feed water, where the overpotential increases requiring more energy while in contrast the hydrogen production rate decreases. These factors do limit the use of PEM commercially as the highly pure water required is not readily available in large quantities.

The production rate of the three electrolyzers was modelled and compared for a run time of 15 minutes to see the amount of hydrogen produced and for energy analysis as provided in Fig. 11.

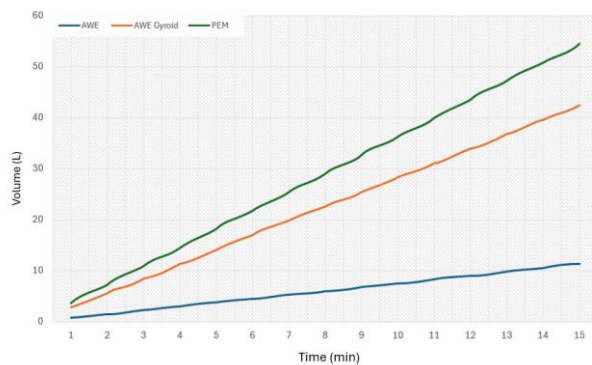


Fig. 11 Electrolyser Hydrogen production

The volume of hydrogen produced for the electrolyzers is in litter, the time measured in minutes, and the gas produced is in STP (standard temperature and pressure). Hydrogen has a molar volume of 22.4L/mol at STP which combined with the time and energy input for each electrolyser gives an estimate of the energy required per mole of Hydrogen produced for each electrolyser.

3.4 Hydrogen utilization

Hydrogen as a combustible fuel has much potential as it has a high energy density, as well as its high-octane number of 130 which is the same as diesel fuel. Energy recovery via hydrogen combustion has two avenues, the direct heat energy recovered from the combustion of hydrogen and the energy recovered through mechanical work from hydrogen combustion, both have their applications. However, the electrical generation from hydrogen combustion will not be considered as a

thermodynamically highly non efficient way, due to the multistep energy conversion from chemical to mechanical to electrical. Where each conversion process increases energy wastage. The bond energy of H-H bonds in hydrogen gas is approximately 432 kJ/mol, the energy released in a combustion reaction is directly proportional to the difference in bond energies between the reactants and the products, for hydrogen combustion the reactants are hydrogen and oxygen, where the oxygen double bond has a bond energy of 495 kJ/mol. As the products of this reaction is only water, which has two O-H bonds each having an average bond energy of 467 kJ/mol. These bond energies and other common chemical bond energies have been experimentally studied and are listed in Table 2 (Song, K., 2019).

Table 2: Molecular bond energy

Single Bonds				Multiple Bonds	
H-H	432	N-H	391	I-I	149
H-F	565	N-N	160	I-Cl	208
H-Cl	427	N-F	272	I-Br	175
H-Br	363	N-Cl	200		
H-I	295	N-Br	243	S-H	347
		N-O	201	S-F	327
C-H	413	O-H	467	S-Cl	253
C-C	347	O-O	146	S-Br	218
C-N	305	O-F	190	S-S	266
C-O	358	O-Cl	203		
C-F	485	O-I	234	Si-Si	340
C-Cl	339			Si-H	393
C-Br	276	F-F	154	Si-C	360
C-I	240	F-Cl	253	Si-O	452
C-S	259	F-Br	237		
		Cl-Cl	239		
		Cl-Br	218		
		Br-Br	193		
				C=C	614
				C≡C	839
				O=O	495
				C=O*	745
				C=O	1072
				N=O	607
				N=N	418
				N≡N	941
				C≡N	891
				C=N	615

Studies indicate Hydrogen has a high flammability as well which is between 4%-75% fuel to air fraction in air under STP (Brun, K. and Allison, T.C., 2022). The flammability test is done using a 1.2m tube with mixture ignited with varying mixtures to determine the flammability. The bond energy analysis provides a comprehensive framework for the energy released during hydrogen combustion, when converting this to useful mechanical energy H₂-ICEs (hydrogen Internal combustion engines) are the best fit as they can make use of the highly combustible properties and convert to mechanical work. (Gao et al. 2025) has studied the efficiency of a H₂-ICE both with a methane mixture and pure hydrogen operation. In only hydrogen internal combustion, the ratio of air in the mixture affects the flammability and combustion pressure, where it was found that a ratio of 0.6 gave the highest peak in flame propagation. The expansion of the combustion is the key defining factor in the generation of mechanical work which depends on the expansion ratio and the temperature inside the ignition cylinder, Fig. 12 provides the efficiency that can be obtained at various expansion ratios and the power generated.

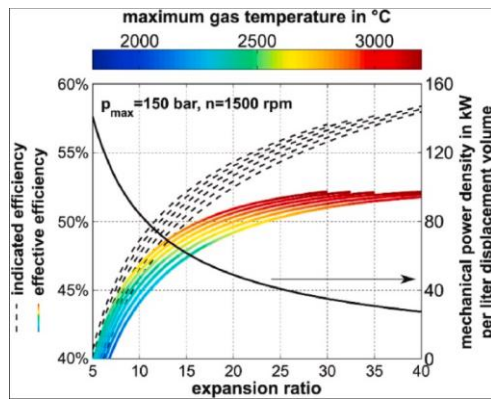


Fig. 12: Mechanical power in respect to expansion ratio (Song, K., 2019)

PEMFC (polymer electrolyte membrane fuel cell) operates a bit similarly to PEM electrolyzers but in reverse where Graphite electrodes are coated with a catalytic material Pt or Pd typically, with a surface area of 50 cm² where the hydrogen gas is dissociated into protons and electrons, the H⁺ ions sent through a PTFE (polytetrafluoroethylene) membrane to be combined with Oxygen gas to be combined into water, while the electrons are diverted through an external circuit and used to produce electricity (Kumar, R. and Subramanian, K.A., 2022). The performance of PEMFCs are dependent on a variety of operational conditions such as temperature, gas flow, oxygen to air ratio and relative humidity. The optimal values for these were considered to produce the best power output.

Table 3: PEMFC optimal parameters

PEMFC Parameter	Optimal value
Temperature (K)	323
Hydrogen flow rate(mol/L)	1
Oxygen - air ratio	0.5
Relative humidity	50%

The optimal operating temperature is 50°C as the higher temperature reduces the activation energy required. To mitigate unnecessary strain on the membrane the gas flow should be equal to half the diffusion rate of protons through the membrane which is dependent on the surface area of the membrane. For the air inflow to reduce the ions into water regular air enrichment is needed with oxygen, between 45%-50% enrichment by volume gives the best performance as it would be expected theoretically as water is final product.

4. Results and Discussion

The results and discussion section analyses and break down of the energy efficiency of green hydrogen production and usage, taking into account the varying factors in green hydrogen that create energy wastage and finding the optimal energy efficiency route for sustainable hydrogen energy production through solar powered electrolysis. The factors considered was the best sustainable energy for electrolysis, hydrogen production methods through electrolysis, energy production through

hydrogen and the overall efficiency of green hydrogen. For these factors their performance was dependent on certain variables, and their effect was taken into account along with improvements that can be made.

4.1 Sustainable energy for Electrolysis

Here it was considered the most readily available sustainable energy, which had two candidates wind power and solar energy. Due to solar power being much more abundant, less location dependant and requiring much less set up and infrastructure and maintenance compared to solar power. The typical commercially available solar panels have a power per square meter of nearly 1000W/m². Though this power is absorbed from solar radiation, its conversion rate to electricity is near 15% in commercially available panels. The power generated by (oxfordpv 2020) was 1087W/m² with an electrical conversion rate of 25% this was used as the power input for electrolyzers as a 2V system using these panels could generate a maximum current output of 135A.

4.2 Hydrogen production methods through electrolysis

Two main types of electrolysis studied were AWE (alkaline water electrolysis) and PEM (Proton exchange membrane), as these have significant advantages over other methods which was discussed in the Literature review, mainly their operational conditions that are near standard conditions and they not require high pressure or temperature which increases the energy and cost for production and are the most widely used in hydrogen production giving much more accurate practical data when it comes to analysing the factors effecting electrolysis. The main factors affecting AWE is charge diffusion and ion transfusion. The charge diffusion rate to the electrolyte greatly depends on the active surface area and current density of the electrode. As regular AWE is done through flat electrodes, the surface area is much more limited and for the same power. The gyroid shape was tested as an improvement due to its high surface area for the same amount of material (4:1 surface area ratio in comparison). Ion transfusion within the electrolyte helps to lower the potential required and increase overall reaction rate. For the Ni based electrodes used the best electrolyte solution for better ion transfusion was 30% KOH solution which reduces the power required for electrolysis by 21%. PEM electrolyzers use Pt and Pd electrodes with a Polymer membrane for hydrogen production and requires ultrapure water for optimal hydrogen production, the electrode material has much more catalytic properties requiring less energy for electrolysis.

The energy required to produce one mol of hydrogen gas (approximately 2 grams by weight and 22.4l under STP) for each electrolysis method was compared. Fig.11 shows the time taken for each electrolyser type to produce a specific volume of hydrogen in Litters under STP, extrapolating the time taken to produce 22.4L per each electrolyser and using Eq. (8) the energy required per mole of hydrogen was calculated.

$$(8) E = a \times j \times t \times V$$

Where:

- E energy (J)

- a area of the electrode in (cm^2)
- j is current density of the system (A/cm^2)
- t time in seconds (s)
- V voltage in the system (V)

Table 4: Electrolyser energy usage

Parameter	AWE	AWE Gyroid	PEM
Time (min)	29.78	7.92	6.27
Time (s)	1786.8	475.2	376.2
area(cm^2)	54.094	203.1992	100
Current density j (A/cm^2)	2	2	1
Voltage (v)	2	2	2
Energy (J)	386620.6368	386241.0394	75240
Energy (kJ)	386.6206368	386.2410394	75.24

Table 4 illustrates the power calculated per each electrolyser to produce one mole of hydrogen gas. Both the flat face electrode and the gyroid electrode obtained approximately similar values for energy consumed per mole of hydrogen produced ($386\text{kJ}/\text{mol}$). However, the time required for this production varied substantially, while the gyroid performed similarly (7.29 mins) to the PEM electrolyser (6.27 mins) for the time taken to produce 1 mole of $\text{H}_{2(\text{g})}$, the regular AWE process took more than three (3.76) times required (29.78 mins). The energy consumed by the PEM ($75.24\text{kJ}/\text{mol}$) is much lower compared to the other two processes, as this using the most optimal values for the PEM electrolyser and its electrode material (Pt/Pd) has more catalytic properties compared to Ni-based electrodes this is to be expected.

4.3 Energy utilization of hydrogen

For energy production through hydrogen three main types of energy produced from hydrogen analysed was its thermal and mechanical energy production through combustion and electrical energy production through hydrogen fuel cells. The combustion of one mole of hydrogen has an energy output of $286\text{kJ}/\text{mol}$, the mechanical energy obtained from H_2 -ICE for one mole hydrogen only has a maximum efficiency of 55% energy conversion (Fig.12).

For the electrical energy produced from hydrogen a PEMFC performance was studied under ideal operating conditions, which has a maximum energy production efficiency of 65% for one mole of hydrogen gas with highly oxygen enriched air.

4.4 Overall efficiency of green hydrogen power

Table. 5: Energy analysis matrix

Energy efficiency Matrix			
kJ	Combustion	Mechanical	Electrical
AWE	-100.62	-229.32	-200.72
AWE Gyroid	-100.24	-228.94	-200.34
PEM	210.76	82.06	110.66
%	Combustion	Mechanical	Electrical
AWE	-26%	-59%	-52%
AWE Gyroid	-26%	-59%	-52%
PEM	280%	109%	147%

Table. 5 shows an energy matrix for the total energy comparison between each hydrogen production method energy input and hydrogen energy output through discussed utilization methods. Both AWE processes have identical overall energy efficiency loss, between 26% - 59% energy loss. Energy recovery through combustion has maximum efficiency for all three methods, as direct conversion to heat has minimal losses, electrical energy recovery shows an efficiency loss of 52% due to energy loss due to ohmic resistance and heat in fuel cells, and mechanical energy loss being nearly 60% due to heat loss in the H_2 -ICE cylinders. PEM energy recovery shows an overall net positive in energy recovery between 9% - 180% gain in efficiency, this increase is not a thermodynamically accurate result as there should be loss created; however, this result could be due to the highly optimal conditions considered in theoretical calculations. This does give insight to the fact that PEM electrolyzers do have a much higher energy efficiency than AWE electrolyzers in comparison. The Gyroid shaped electrode did achieve a higher hydrogen production rate equivalent to the PEM in optimal conditions which does make it a viable alternative for improved hydrogen production.

5. Conclusion

This study explores the feasibility of green hydrogen as a viable alternative to commercial energy production methods, and its overall energy efficiency. An in-depth study into the different methods of hydrogen production through electrolysis was conducted, along with the best sustainable energy source to be used and energy produced through hydrogen to find the most efficient path for green hydrogen energy production. Solar energy was the most viable source of energy for electrolysis due to its high global availability and ease of setup compared to other sustainable sources. Three electrolysis methods were considered; optimal conditions were used for each to gain maximum energy efficiency during hydrogen production. AWE was one of the most feasible hydrogen production methods to be used in conjunction with solar power and a gyroid shaped electrode was studied as a viable improvement to typical electrode shape, which showed a significant improvement in the hydrogen production rate (376%) but did have similar energy consumption to the regular electrode. This increase is to be expected theoretically as the gyroid has a much higher surface area making the time taken to produce a mole of hydrogen gas much lower, however since the energy input required to

produce hydrogen does not change its efficiency lies only in the rate of gas produced. This does require further research through CFD modelling to analyse other AWE effecting factors such as gas bubble dissipation that the gyroid shape may have improvements in. PEM electrolysis showed similar hydrogen production rate efficiency to the gyroid while requiring a fractional energy input compared to AWE, which is a factor in relation to its higher catalytic electrode material and highly less ohmic losses. However, these results may vary significantly in real word applications as the conditions are highly optimized for its operation. Further, research into the PTFE membrane and catalytic effect of the electrode material of PEM would be required along with experimental research into PEM performance to gain much more accurate energy analysis of the electrolyser. The best method for hydrogen production commercially would be AWE-gyroid electrolysis as it does give a higher production rate with lower cost of operation. For optimal hydrogen production PEM do have a significant efficiency in production rate and energy used if the under optimal conditions.

The power acquired by hydrogen does support thermodynamic data, as combustion heat generation is the most energy efficient path for any green hydrogen produced but in obtaining mechanical or electrical energy there is energy loss comparatively. For more improvements in these methods research into experimental H₂-ICE and fuel cell performance will be conducted.

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